



EVOLUTION OF MARS ROVERS

What is a Mars rover and why do we need one (or many)?

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ver since Italian astronomer Giovanni Virginio Schiaparelli first mapped the so-called "Canali" on

Mars in 1857, the red planet has captured the imagination of astronomers and the public alike. The curiosity has been engrained in us, and we have sent multiple rovers over the years to explore the surface of the mysterious red planet. Simply put, a rover is a remotely controlled vehicle designed to navigate a terrain. The first Martian landers were Mars 2 and Mars 3 by the Soviet Union in 1971. So far, NASA has successfully sent five rovers to Mars starting with the rover named Sojourner in 1997.

China also launched its own rover named Zhurong in 2021 as part of Tianwen-1 mission. India launched its first interplanetary Mars Orbiter Mission (MOM), also called Mangalyaan in 2014. In case you were

wondering, none of these rovers/orbiters have encountered any form of intelligent life and none were crushed by a Transformer... yet! Each country would have its own interests to explore Mars. These goals are lofty and represent the pinnacle of humankind's scientific endeavor. But first, we must get there in one piece!

THE FIRST ROVERS — UNRAVELING THE PAGES OF MARTIAN HISTORY

Getting to Mars is no easy task, and each mission presents its own set of challenges. In the words of Steve Squyres, Principal Investigator for NASA's Spirit and Opportunity Mars Rovers, "Mars is a spacecraft graveyard".

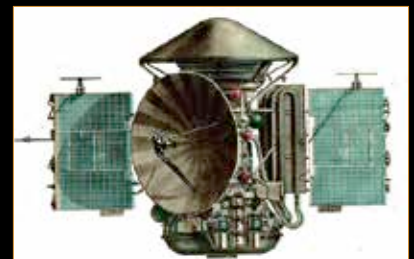
The arduous 150.9 million mile-journey from Earth to Mars takes seven months if the spacecraft travels nearly 97,000 kilometers an hour. Squyres says that hitting the precise landing site on Mars from Earth is like, "shooting a basketball from Los Angeles to New York and having it go through the hoop without touching the rim". That being said, there have been some notable attempts to get the ball rolling on Mars.

Mars 2 and Mars 3

The Mars 2 and Mars 3 missions comprised of identical spacecraft launched by the Soviet Union on May 19 and May 28, 1971, respectively.

Mars 2's lander became the first human-made object to land on Mars. However, the lander crashed on Mars possibly due to a steep angle of entry and was lost. Due to this, the Pro-M rover that the lander carried could not be deployed. Mars 3 was designed to perform a soft landing on Mars and send back data on meteorological conditions, atmospheric composition, and soil properties. Mars 3 was first spacecraft to make a successful soft landing on Mars.

Between December 1971 and March 1972, both Mars 2 and Mars 3 transmitted a total of 60 images. On August 22, 1972, after completing 362 and 20 orbits, respectively, Mars 2 and Mars 3 were declared mission complete.



Sojourner

Sojourner was part of NASA's Mars Pathfinder mission, which was conceived with the objective of delivering a lander and a robotic rover on Mars in a cost-effective manner. The Pathfinder mission was launched on December 4, 1996. Sojourner landed on Mars on July 4, 1997, in an ancient flood plain in the Martian northern hemisphere named Ares Vallis. It sent back more than 550 images while the lander transmitted more than 16,500. The Pathfinder mission helped NASA collect vital information pertaining to rocks at the landing site, indications of liquid water flows in the past, water ice clouds in lower atmosphere, weather patterns, and more.

NASA last heard from Pathfinder on September 27, 1997. The cause of failure is suspected to be a dead battery.